If $y = e^x,\ \dfrac{dy}{dx} = e^x,\ \dfrac{d^2y}{dx^2} = e^x$

$$(x-1)e^x - xe^x + e^x = 0$$

Let $y = ue^x$

$$\frac{dy}{dx} = e^x\left(u + \frac{du}{dx}\right)$$

$$\frac{d^2y}{dx^2} = e^x\left(u + 2\frac{du}{dx} + \frac{d^2u}{dx^2}\right)$$

$$\begin{aligned} 0 &= (x-1)e^x\left(u + 2\frac{du}{dx} + \frac{d^2u}{dx^2}\right) - xe^x\left(u + \frac{du}{dx}\right) + ue^x \\ &= e^x\left((x-1)\frac{d^2u}{dx^2} + (x-2)\frac{du}{dx}\right) \end{aligned}$$

$$\because e^x > 0$$

$$\therefore (x-1)\frac{d^2u}{dx^2} + (x-2)\frac{du}{dx} = 0$$

Let $\dfrac{du}{dx} = v$

$$(x-1)\frac{dv}{dx} + (x-2)v = 0$$

$$\frac{1}{v}\frac{dv}{dx} = -\frac{x-2}{x-1} = -1 + \frac{1}{x-1}$$

$$\ln v = -x + \ln|x-1| + C$$

$$v = A(x-1)e^{-x}$$

$$\frac{du}{dx} = A(x-1)e^{-x}$$

$$u = -Axe^{-x} + B$$

$$\implies y = ue^x = -Ax + Be^x = Ax + Be^x$$

This solution was submitted by **Jon**
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